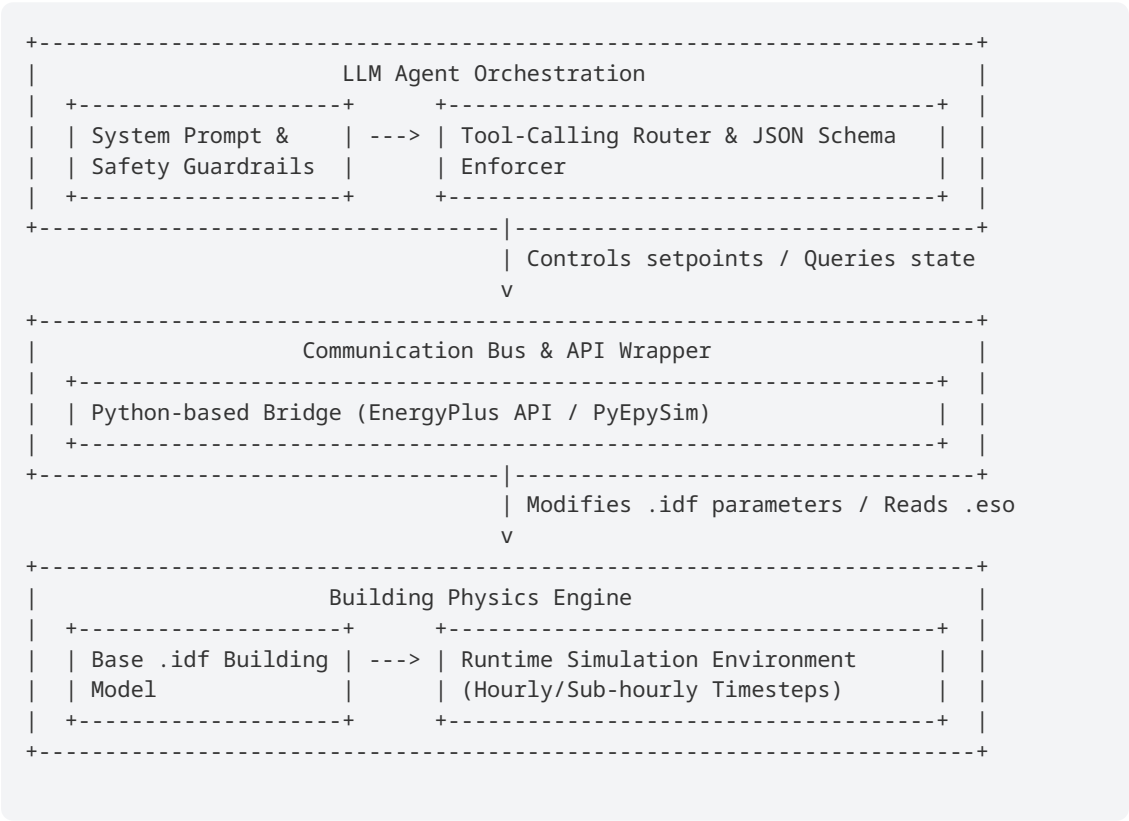


System Architecture Document: Eco-Loop Building Agents

1. Executive Summary

Eco-Loop Building Agents is an autonomous, closed-loop framework designed to minimize commercial and residential building energy consumption (kWh) via Large Language Model (LLM) agent orchestration while strictly preserving indoor thermal comfort bounds. The system bridges modern AI orchestration layers with industry-standard building physics engines (EnergyPlus).

2. High-Level System Architecture



3. Core Component Breakdown

3.1. Tool-Calling Architecture

The orchestration engine uses structured function calling to interface with the simulation environment:

- **State Acquisition Tool (`get_building_telemetry`):** Fetches current zone air temperature, mean radiant temperature, relative humidity, outdoor ambient temperature, and instantaneous HVAC power draw (kW).
- **Action Execution Tool (`set_hvac_parameters`):** Dynamically adjusts thermostat heating/cooling setpoints or schedule profiles for the upcoming simulation timestep.

3.2. Prompt Engineering Strategies

To ensure reliable, physically safe, and energy-efficient outputs, the LLM prompt employs:

- **Role-Based Persona Conditioning:** *"You are an elite Building Energy Systems Engineer specializing in HVAC optimization and thermal comfort compliance under ASHRAE Standard 55."*
- **Hard Operational Guardrails:** Explicit instructions setting absolute boundaries (e.g., Indoor temperatures must never drop below 20°C or exceed 26°C).
- **Structured Output Enforcement:** Forcing responses via JSON schema constraints to eliminate parsing errors and guarantee programmatic execution.

3.3. Prompt Latency Management

To handle real-time or accelerated building simulation loops without timing bottlenecks:

- **State Compression:** Instead of passing full raw text histories, historical telemetry is condensed into a rolling vector array representing the last N timesteps of thermal drift and energy consumption trends.
- **Token Budgeting & Low-Latency Configurations:** System prompts and context payloads are tightly trimmed to minimize inference time. Asynchronous execution handles background metric logging while awaiting API model responses.

3.4. Handling Lengthy Simulation Logs

EnergyPlus produces massive output artifacts (`.eso`, `.csv`, `.out`), which easily saturate LLM context windows. This is solved via a **Data Reduction Pipeline**:

- **Sliding-Window Parsing:** A custom Python log-parser extracts only key performance indicators (KPIs) per timestep (mean temperature, total kWh delta, peak demand, and constraint violation counts).
- **Storage Offloading:** Raw files remain saved locally in the `results/` directory for final visualization, while only high-level statistical rollups are fed back into the agent's active memory context.